ADAPTIVE DIGITAL SAT PRACTICE TEST

DSAT_Mar_2025_INT_v4_W

Full-Length Practice Examination

Section 1: Reading & Writing — Module 1 (27 Qs) | Module 2 Easier (27 Qs) | Module 2 Harder (27 Qs)

Section 2: Math — Module 1 (22 Qs) | Module 2 Easier (22 Qs) | Module 2 Harder (22 Qs)

Total Questions: 147

Section 1: Reading & Writing — Module 1 (27 Questions)

Question 1

Skill: Words in Context | Difficulty: Easy

In several of her ceramic works, Maya Lin employs organic, flowing forms to represent natural landscapes rather than rendering them with photographic precision. For example, her 2009 installation *Wave Field* is highly stylized and therefore stands apart from some of her other pieces in which the viewer can readily ______ recognizable terrain.

Which choice completes the text with the most logical and precise word or phrase?

A) detect
B) recall
C) dismiss
D) classify

Question 2

Skill: Words in Context | Difficulty: Easy

During the early decades of commercial radio, station managers had a strong incentive to attract large audiences. As media historian Kenji Watanabe argues, some broadcasters turned to music critics to shape the public's ______ the emerging medium. Critics who highlighted parallels between radio programs and established concert traditions could help legitimize broadcasting as a serious cultural outlet.

Which choice completes the text with the most logical and precise word or phrase?

A) contribution to
B) appreciation of
C) imitation of
D) dependence on

Question 3

Skill: Words in Context | Difficulty: Medium

The botanical illustrations housed in institutions such as the Royal Botanic Gardens at Kew (which holds an extensive archive of watercolor paintings) are celebrated for the ______ of the artistry behind them — the illustrators consulted living specimens, dried herbarium sheets, and expert taxonomists to guarantee the fidelity of every petal and leaf vein.

Which choice completes the text with the most logical and precise word or phrase?

A) meticulousness
B) obscurity
C) brevity
D) spontaneity

Question 4

Skill: Words in Context | Difficulty: Medium

Innovation clusters develop when numerous enterprises in complementary sectors ______ a single region, as with semiconductor designers and fabrication-equipment manufacturers in the Hsinchu Science Park in Taiwan. Scholars have long assumed that firms congregate for identical reasons, but Annika Bergström and colleagues demonstrated that the drivers of clustering in certain industries are only loosely associated with clustering in others.

Which choice completes the text with the most logical and precise word or phrase?

- A) settle in
- B) defer to
- C) compete with
- D) agree on

Question 5

Skill: Words in Context | Difficulty: Medium

Contemporary concert halls are routinely outfitted with sophisticated acoustic panels and lighting rigs to immerse the audience in a performer's sonic world. Because concertgoers have become accustomed to meticulously engineered environments, performances held in sparse, unadorned venues can catch audiences off guard. But minimalist staging was likely ______ audiences centuries ago: elaborate acoustic treatments were uncommon before the 1800s.

Which choice completes the text with the most logical and precise word or phrase?

- A) resented by
- B) puzzling to
- C) familiar to
- D) thrilling for

Question 6

Skill: Words in Context | Difficulty: Hard

Coral reef monitoring programs across the Indo-Pacific frequently rely on volunteer divers whose species-identification abilities vary considerably. This inconsistency can ______ the reliability of biodiversity surveys, because records submitted by inexperienced observers may conflate morphologically similar species, inflating apparent richness at some sites while underreporting it at others.

Which choice completes the text with the most logical and precise word or phrase?

- A) undermine
- B) confirm
- C) advertise
- D) predict

Question 7

Skill: Text Structure and Purpose | Difficulty: Medium

The following passage is adapted from Willa Cather's 1905 short story "The Sculptor's Funeral." Martin Hargrove is a prominent citizen in a small Kansas town.

By the residents of Sand City and its surrounding farmsteads, Mr. Hargrove was esteemed as a man of uncommon shrewdness and practical judgment, one whose advice on matters of land and livestock carried the weight of settled law. He never ventured an opinion without first weighing every foreseeable consequence; he listened more than he spoke and spoke only when he was certain his words would serve him; he took care that neither friend nor stranger could discern the calculations turning behind his placid expression, and he never furnished anyone a reason to question his goodwill.

Which choice best describes the overall structure of the text?

- A) It establishes that Mr. Hargrove was once well regarded by his neighbors and then recounts the incidents that diminished their esteem.
- B) It contrasts Mr. Hargrove's public demeanor with his private ambitions and then explains why he worked to conceal one from the other.
- C) It suggests that Mr. Hargrove's neighbors viewed him more favorably than city dwellers did and then specifies why his rural neighbors were so easily deceived.
- D) It presents the positive reputation Mr. Hargrove enjoyed among his neighbors and then describes the behaviors that allowed him to sustain that reputation.

Question 8

Skill: Text Structure and Purpose | Difficulty: Hard

Researchers Tomoko Ishida and colleagues have studied how parallel phenotypic adaptation — a phenomenon in which organisms in separate populations independently develop similar traits — can arise through shared genetic pathways. Separately, Hanna Kokko and collaborators have examined how such adaptation occurs via distinct genetic routes, but the relative contribution of shared versus distinct pathways remains poorly characterized. This gap motivated geneticists Priya Ramanathan and Luis Delgado to assess both routes of parallel adaptation within a single controlled experiment for their 2018 paper.

Which choice best states the overall structure of the text?

- A) It outlines a broadly held assumption about a biological process, describes two experiments that challenged that assumption, and explains why the assumption persists.
- B) It introduces a particular experiment, explains how two earlier investigations approached the same process differently, and critiques the first experiment's methodology.
- C) It surveys the history of a research field, identifies three landmark studies that advanced the field, and speculates about future directions.
- D) It describes a phenomenon, provides examples of two studies that have each examined the phenomenon from a different angle, and mentions a third study of that phenomenon.

Question 9

Skill: Central Ideas and Details | Difficulty: Medium

The following text is from a translation of Fumiko Enchi's 1957 novel *The Waiting Years*. The narrator, Tomo, has just arrived at a rented house in a coastal town.

Through the years there had been countless occasions when fortune had handed her sudden reversals — losses she had not anticipated, alliances that dissolved without warning, relocations forced upon her by circumstances she could neither predict nor prevent. Sometimes she had met these upheavals with composure; more often she had not. Yet never before had she been so conscious of standing at a turning point as she was that rainy evening in November when she slid open the front door and heard the hollow echo of her footsteps in the bare rooms. What lay behind her was a tangled history, and what stretched ahead was an expanse of unmarked time she would somehow have to fill.

Which choice best states the main idea of the text?

- A) The narrator is enthusiastic about the financial opportunities that a new location will provide.
- B) The narrator senses that a specific moment represents a significant transition in her life.
- C) The narrator has consistently relied on a careful strategy for managing life's disruptions.
- D) The narrator wishes to acknowledge the people who have supported her through hardship.

Question 10

Skill: Command of Evidence (Quantitative) | Difficulty: Medium

In India, reliance on traditional biomass (such as dung, crop residues, and firewood) as a share of total household energy consumption declined by roughly 55 percent between 2001 and 2021. Analysts typically attribute such transitions to the energy staircase, a framework positing that fuel choice is driven primarily by household wealth (specifically, that cleaner fuels replace traditional biomass as earnings grow). Ritu Sharma and Arvind Patel's examination of energy use in rural Gujarat, however, reveals a more complex picture: household fuel adoption was uneven, context-dependent, and shaped by multiple variables, including the availability of subsidized LPG cylinders and regional cooking customs.

Based on the text, how would a proponent of the energy staircase most likely explain the change observed in India?

- A) Between 2001 and 2021, household earnings rose in India, leading a larger share of families to meet their energy needs with cleaner fuels.
- B) Starting in 2001, subsidized LPG cylinders became more widely distributed in rural India, enabling households at every income level to abandon traditional biomass.
- C) Traditional biomass use fell in India between 2001 and 2021 because economic changes during that period made biomass more costly relative to household budgets.
- D) Household earnings in India increased from 2001 to 2021, which diminished the link between income and household fuel selection.

Question 11

Skill: Command of Evidence (Quantitative) | Difficulty: Medium

While compiling data for a report on global tin production, a student finds statistics on tin output in several countries for 2000 and 2022.

[PLACEHOLDER: Table titled 'Thousands of Metric Tons of Tin Produced in 2000 and 2022' with columns: Country | 2000 | 2022. Rows: Bolivia | 12 | 24; China | 99 | 68; Indonesia | 52 | 58; Myanmar | 5 | 31; Nigeria | 2 | 7. The passage before the blank reads: 'The student notes that Indonesia produced 52 thousand metric tons of tin in 2000 and ______']

Which choice most effectively uses data from the table to complete the statement?

A) 58,000 metric tons of tin in 2022.
B) 24,000 metric tons of tin in 2022.
C) 73,000 metric tons of tin in 2022.
D) 31,000 metric tons of tin in 2022.

Question 12

Skill: Command of Evidence (Textual) | Difficulty: Hard

"Heritage Quilts" is a 1994 poem by Mariama Jeffries. In the poem, the speaker suggests that the quilts described in the poem are meaningful because they were crafted from the quiltmaker's own lived experience, noting that ______

Which choice most effectively uses a quotation from 'Heritage Quilts' to illustrate the claim?

A) a young observer "understands that Grandma Rose / Never found her patterns in any catalogue at all. / Instead they grew / Straight from the soil of her own days."
B) the figures in the quilts "Weave themselves gently / Into the rhythm of old Grandma Rose's hands, / Weave themselves gently."
C) faded threads "loop and tangle / Across Grandma Rose's quilts."
D) the quilts are stitched during "Autumn evenings by the kitchen window."

Question 13

Skill: Command of Evidence (Quantitative) | Difficulty: Hard

Researchers examined how appreciation of a novel is influenced when its ending has been revealed in advance (when the reader has prior knowledge of a critical plot twist). As part of the study, participants rated their appreciation of one novel whose ending was revealed beforehand and one novel whose ending was not revealed. For every novel, participants who knew the ending in advance reported greater appreciation than those who did not. However, the magnitude of this difference varied across the novels, as is best illustrated by the appreciation ratings for ______

[PLACEHOLDER: Bar chart titled 'Novel Appreciation: Revealed vs. Unrevealed Endings.' X-axis lists six novels: 'A Fading Star,' 'The Glass Garden,' 'The Last Lantern,' 'Ironside,' 'Ember Road,' 'Windfall.' Y-axis: 'Average appreciation rating (1 = lowest; 10 = highest)' from 0 to 8. Each novel has two bars — light gray (unrevealed) and dark gray (revealed). Key data: 'A Fading Star' unrevealed ≈ 3.5, revealed ≈ 5; 'The Glass Garden' unrevealed ≈ 5, revealed ≈ 5.5; 'The Last Lantern' unrevealed ≈ 6.5, revealed ≈ 7.5; 'Ironside' unrevealed ≈ 6.5, revealed ≈ 7; 'Ember Road' unrevealed ≈ 5, revealed ≈ 5.5; 'Windfall' unrevealed ≈ 5, revealed ≈ 5.5. The largest gap between bars is for 'A Fading Star' (≈ 1.5) and the smallest gap is for 'The Glass Garden' (≈ 0.5).]

Which choice most effectively uses data from the graph to complete the statement?

- A) "The Glass Garden" and "Ironside."
- B) "Ember Road" and "Windfall."
- C) "The Last Lantern" and "Ironside."
- D) "The Last Lantern" and "Windfall."

Question 14

Skill: Inferences | Difficulty: Hard

Although Western Atlantic loggerhead sea turtles generally alternate between their nesting beaches along the southeastern United States and their open-ocean foraging grounds in the Sargasso Sea, a subset of this population — designated the Mid-Atlantic Bight (MAB) group — feeds along the continental shelf from Virginia to Massachusetts instead. Notably, individuals in the MAB group attain smaller adult sizes than other Western Atlantic loggerheads, despite exhibiting comparable juvenile growth rates. Researchers hypothesize that this difference may reflect an adaptation to the distinct prey assemblages available in the MAB foraging range.

Which finding, if true, would most directly support the researchers' claim regarding the size of MAB loggerheads?

- A) When foraging along the continental shelf from Virginia to Massachusetts, MAB turtles are in closer proximity to commercial shipping lanes than Western Atlantic loggerheads in the main group are when foraging in the Sargasso Sea.
- B) The mean carapace length of MAB loggerheads observed along the mid-Atlantic coast has remained relatively stable over recent decades, while the mean carapace length of loggerheads in the main group has slightly declined.
- C) When present along the mid-Atlantic coast, MAB loggerheads tend to feed on benthic invertebrates at depths inaccessible to turtles as large as those in the main group.
- D) Certain crustacean prey species available along the mid-Atlantic coast where MAB loggerheads feed are absent from the Sargasso Sea waters where loggerheads in the main group forage.

Question 15

Skill: Inferences | Difficulty: Hard

Chimamanda Ngozi Adichie's 2013 novel *Americanah* is frequently characterized as semi-autobiographical. That label has some justification — there are notable similarities between the experiences of the novel's protagonist, Ifemelu, and those of Adichie — but it should not be taken to imply that every character and incident depicted in *Americanah* is drawn from life. The novel is substantially a work of imagination, and readers who overlook this fact and instead attempt to map more and more biographical parallels therefore risk ______

Which choice most logically completes the text?

- A) imposing unwarranted correspondences between *Americanah* and Adichie's biography.
- B) downplaying the degree to which Adichie incorporated real-world material when writing *Americanah*.
- C) overstating *Americanah*'s readership relative to its actual popularity.
- D) exaggerating the influence that earlier novelists had on Adichie's literary style.

Question 16

Skill: Standard English Conventions | Difficulty: Easy

The earliest public aquarium in Europe opened during the height of the Victorian fascination with natural history in 1853. Situated in Regent's Park, London, the facility was called the Fish House at the London Zoo. It was stocked with freshwater and marine species collected ______ rivers and coastal waters of the British Isles.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) from;
- B) from
- C) from:
- D) from,

Question 17

Skill: Standard English Conventions | Difficulty: Easy

The Museo de Arte Popular in Mexico City, a cultural institution housed in a renovated Art Deco building, is one of the foremost repositories of folk art in Latin America. The museum's permanent galleries ______ visitors about a wide array of traditional crafts, from lacquerware and pottery to textile weaving and papier-mâché sculpture.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) having educated
- B) educating
- C) to educate
- D) educate

Question 18

Skill: Standard English Conventions | Difficulty: Easy

In April of 1964, the single "My Guy" by Mary Wells became a chart sensation. After holding a position in the top five for several weeks, ______ reached No. 1 on the Billboard Hot 100 list of most popular recordings.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) those
- B) they
- C) each
- D) it

Question 19

Skill: Standard English Conventions | Difficulty: Medium

An employee-owned enterprise is a company in which the workforce collectively holds a controlling ownership stake. This arrangement differs from conventional corporate structures in which a small number of outside shareholders governs the firm. Because the revenue generated by an employee-owned enterprise is distributed among all staff members — who are simultaneously owners — the employees ______ directly from its profitability.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) gain
- B) had gained
- C) gained
- D) were gaining

Question 20

Skill: Standard English Conventions | Difficulty: Medium

The 2018 documentary *Conscience of the Coast* was directed by Paloma Reyes. It chronicles how Indigenous Oaxacan traditions informed the environmental conservation movement along Mexico's Pacific coastline in the late twentieth ______ how the ecological stewardship practices Indigenous women maintained in their communities shaped the first marine protected areas in the region.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) century revealing
- B) century; revealing
- C) century. Revealing
- D) century, revealing

Question 21

Skill: Transitions | Difficulty: Medium

In general, narrow-bodied watercraft are more hydrodynamic than wide-bodied ones. For instance, the tapered hull of a competitive rowing shell allows it to slice through water with minimal drag. _______ a flat-bottomed barge encounters considerably more water resistance, reducing its speed.

Which choice completes the text with the most logical transition?

A) As a result,
B) Specifically,
C) In summary,
D) On the other hand,

Question 22

Skill: Transitions | Difficulty: Medium

In planetary science, a ring of particles orbiting a large body within the body's Roche limit is expected to remain dispersed as a ring, whereas a ring of particles orbiting beyond that limit would likely coalesce into a moon. Bruno Morgado and colleagues, _______ observed a dense band of material circling the trans-Neptunian object Quaoar at a distance of 2,500 miles, well outside the computed Roche limit of 1,100 miles, that has nevertheless stayed intact.

Which choice completes the text with the most logical transition?

A) predictably,
B) for instance,
C) similarly,
D) however,

Question 23

Skill: Rhetorical Synthesis | Difficulty: Medium

- Zacatecas is a state in north-central Mexico.
- States are federal entities responsible for delivering many public services to their populations.
- One service they deliver is emergency response.
- Zacatecas encompasses an area of approximately 75,040 km².
- Mexico is divided into 32 federal entities.

The student wants to emphasize the geographic extent of Zacatecas. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Zacatecas is one of 32 federal entities, referred to as states, across Mexico.
B) Delivering emergency services is just one example of the public responsibilities that states fulfill.
C) Zacatecas — a state in north-central Mexico — delivers a broad range of public services to its population.
D) The state of Zacatecas in north-central Mexico encompasses an area of approximately 75,040 km².

Question 24

Skill: Rhetorical Synthesis | Difficulty: Medium

- Most fungal and lichen species on the Faroe Islands, a North Atlantic archipelago, are non-native.
- In a 2021 survey, researchers wanted to determine what role non-native fungi play in nutrient cycling on the islands.
- Researchers catalogued fungal specimens found in soil samples from non-native woodland plots.
- *Cladonia rangiferina*, a lichen, was one of twelve native species catalogued.
- *Trametes versicolor*, a bracket fungus, was one of thirty-four non-native species catalogued.
- Researchers concluded that non-native fungi play a significant role in nutrient cycling on the islands.

The student wants to emphasize a contrast between the two fungi. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Most fungal and lichen species catalogued on the Faroe Islands are non-native.

B) Although *Cladonia rangiferina* and *Trametes versicolor* can both be found on the Faroe Islands, only the former is native to the archipelago.

C) A 2021 survey catalogued fungal specimens collected from soil samples by researchers on the Faroe Islands to assess what role non-native fungi play in nutrient cycling.

D) Specimens of *Cladonia rangiferina* and *Trametes versicolor* were found in the soil samples of non-native woodland plots on the Faroe Islands, according to a 2021 survey.

Question 25

Skill: Rhetorical Synthesis | Difficulty: Hard

- A supercontinent is a single landmass composed of most or all of Earth's continental crust.
- Over geologic time, continental plates converge to form supercontinents, which later fragment.
- This cycle is believed to span hundreds of millions of years and is known as the supercontinent cycle.
- Rodinia was a supercontinent that assembled about 1.1 billion years ago.
- Gondwana was a supercontinent that assembled about 550 million years ago.

The student wants to emphasize the chronological relationship between the formation of the two supercontinents. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Rodinia assembled roughly 1.1 billion years ago but eventually fragmented.

B) Rodinia and Gondwana were both vast landmasses composed of most or all of Earth's continental crust.

C) The supercontinent Gondwana coalesced hundreds of millions of years after the supercontinent Rodinia.

D) Assembling and fragmenting over hundreds of millions of years, supercontinents consist of most or all of Earth's continental crust.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Hard

- Generally, a material will weaken when subjected to repeated mechanical stress.
- The repeated application of mechanical stress to a material is known as cyclic loading.
- A 2021 study led by Fiona Hartley and Omar El-Amin tested the fatigue behavior of various textile fibers.
- When a sample of natural silk fiber was cyclically loaded, its tensile strength decreased by 18%.
- When a sample of synthetic nylon filament was cyclically loaded, its tensile strength decreased by 27%.

The student wants to highlight a similarity between silk fibers and nylon filaments. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In 2021, two research groups measured the effects of cyclic loading on different textile fibers, including silk and nylon.

B) Both silk fibers and nylon filaments lose tensile strength when subjected to cyclic loading, according to a 2021 study.

C) Investigators found that when the fibers were cyclically loaded, the reduction in tensile strength of nylon was greater than that of silk.

D) Subjecting a material to repeated mechanical stress will generally weaken it, a process known as fatigue.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Hard

- In a 2005 study, Nakamura and Chen tested the effect of organic mulch on seed germination in a woodland setting.
- The test site was a temperate woodland with acidic sandy soil in southern Japan.
- The researchers found that in these environmental conditions the presence of organic mulch had a positive effect on germination.
- Germination is the process by which a seed sprouts and begins developing into a young plant.

Which choice most effectively uses information from the given sentences to present the study's findings to an audience already familiar with the concept of germination?

A) The results of Nakamura and Chen's study were published in 2005.

B) In a 2005 study by Nakamura and Chen, organic mulch was found to have a beneficial effect on germination, which is the process by which a seed develops into a young plant.

C) The influence of organic mulch, which consists of decomposed bark and other plant debris, on germination has been the subject of horticultural research.

D) Nakamura and Chen found that in a temperate woodland with acidic sandy soil, the presence of organic mulch had a beneficial effect on germination.

Section 1: Reading & Writing — Module 2, Easier Path (27 Questions)

Question 1

Skill: Words in Context | Difficulty: Easy

Like other coastal estuaries, the Mekong River delta is ______: it is a perpetually shifting mosaic of waterways and sandbars that alter their contours as the river carries fresh alluvial deposits to the point where its current meets the tides of the South China Sea.

Which choice completes the text with the most logical and precise word or phrase?

- A) volatile
- B) unchanging
- C) matchless
- D) resilient

Question 2

Skill: Words in Context | Difficulty: Easy

Aesop of Thrace (present-day northeastern Greece) was a semi-legendary storyteller of the sixth century BCE who primarily composed moralizing fables about animals. His tale *The Tortoise and the Hare*, however, is ______: featuring themes that anticipate modern sports psychology (such as overconfidence and pacing strategy), it is considered by some scholars as the earliest recognizable work in the genre.

Which choice completes the text with the most logical and precise word or phrase?

- A) conventional
- B) obscure
- C) pioneering
- D) impartial

Question 3

Skill: Words in Context | Difficulty: Easy

The tapestry collection at the Musée de Cluny in Paris (which includes a renowned series depicting a noblewoman with a unicorn) is admired for the ______ of its preservation — the curatorial staff have maintained carefully regulated humidity, lighting, and temperature to ensure that the medieval dyes retain their vibrancy across the centuries.

Which choice completes the text with the most logical and precise word or phrase?

- A) thoroughness
- B) mystery
- C) simplicity
- D) originality

Question 4

Skill: Central Ideas and Details | Difficulty: Easy

Established in San Antonio, Texas, in 2001, the Museo Alameda was the first museum affiliated with the Smithsonian Institution dedicated to presenting Latino art and culture. Although it closed its physical galleries in 2012, its digital archive of more than 2,000 items remains accessible. A separate institution, the National Museum of the American Latino, was authorized by Congress in 2020 and is planned for the National Mall in Washington, D.C.; it will focus on the broad sweep of Latino contributions to the United States.

Which choice best states the main purpose of the text?

A) To describe the founding of two institutions, including how each secured its exhibition space.
B) To compare the sizes of two institutions' permanent collections.
C) To provide information about two institutions, including each institution's focus.
D) To outline a historical trend that led to the creation of two institutions.

Question 5

Skill: Text Structure and Purpose | Difficulty: Medium

Sculptures by the Oaxacan woodcarvers — a loosely connected network of artisan families active in the Valles Centrales of Oaxaca since the mid-twentieth century — are identifiable by their shared reliance on the same general motifs and materials: copal wood, aniline dyes, and fantastical animal forms known as *alebrijes*. Yet within this common vocabulary there was room for individual expression: Manuel Jiménez's *Dragón Rojo*, for example, may at first glance resemble other Oaxacan carvings, but it distinguishes itself through its finer detailing and more naturalistic proportions.

Which choice best describes the overall structure of the text?

A) It describes a shared artistic tradition of a group of artisans and then illustrates how one individual introduced distinctive variations within that tradition.
B) It explains a widespread belief that a particular group of artisans produced unoriginal work and then supplies specific evidence that reinforces that belief.
C) It provides the historical background that explains a group of artisans' collective style and then identifies the moment at which several members began experimenting with unconventional themes.
D) It recounts how a particular group of artisans began collaborating and then traces how one member became especially influential among them.

Question 6

Skill: Cross-Text Connections | Difficulty: Medium

Text 1

Subsurface ocean worlds are a category of moons and dwarf planets that may harbor liquid water beneath icy crusts — a condition potentially hospitable to microbial life. Thermal models indicate that for candidate subsurface oceans, the range of the habitable zone (HZ), the orbital distance from a host planet that permits internal heating sufficient to maintain liquid water, extends inward to roughly 5 planetary radii. In 2019, Keiko Tanaka et al. identified Enceladus, a moon of Saturn, as a strong subsurface-ocean candidate, noting that the moon orbits at approximately 4 planetary radii from Saturn — near the inner boundary of the HZ.

Text 2

In a 2022 study, Dmitri Volkov et al. argued that the icy crusts of Enceladus and similar moons absorb more tidal energy than earlier models assumed, causing elevated internal temperatures and accelerated ice-shell thinning. Unlike prior estimates, Volkov et al.'s calculations therefore pushed the inner edge of these moons' HZ outward to as far as 7 planetary radii.

Based on the texts, how would Volkov et al. (Text 2) most likely respond to Tanaka et al.'s research, as presented in Text 1?

A) By noting that unlike the subsurface-ocean candidate Tanaka et al. studied, most other moons with icy crusts are probably situated within the HZ.

B) By arguing that Tanaka et al. used a model whose estimates of internal heating on subsurface-ocean candidates are likely too conservative.

C) By contending that Enceladus and similar moons are unlikely to maintain liquid water because they orbit too far from their host planets.

D) By asserting that the chemical makeup of the ice shell on the moon Tanaka et al. identified suggests that this moon's subsurface is unlikely to contain liquid water.

Question 7

Skill: Inferences | Difficulty: Medium

Mexican architect Pedro Ramírez Vázquez's prolific career, which extended from the 1950s to the 1990s, progressed through distinct stylistic phases. After studying pre-Columbian building practices and traveling extensively through Europe in the early 1960s, Ramírez Vázquez began synthesizing principles drawn from Brutalism and vernacular Mexican construction in his work, as evidenced in the Museo Nacional de Antropología, whose monolithic concrete canopy contrasts with his earlier projects in Monterrey, such as the Edificio Garza Sada, which displays the ornamental motifs and warm masonry typical of regional Neocolonial design.

Information in the text best supports which statement about the design of the Edificio Garza Sada?

A) It represents a transitional phase between the early and late periods of Ramírez Vázquez's career.

B) It reflects an ornamental and regional approach that Ramírez Vázquez later moved away from.

C) It shows the influence of Ramírez Vázquez's exposure to European Brutalism in the 1960s.

D) It is characteristic of the Monterrey architectural tradition that shaped Ramírez Vázquez's entire career.

Question 8

Skill: Inferences | Difficulty: Hard

Charleston, South Carolina, has reinforced engineered barriers along 48% of its harbor shoreline to protect infrastructure from storm surges and tidal flooding, a practice known as shoreline armoring. To compare the effects of two categories of armoring structures — seawalls and revetments — on tidal marsh invertebrate communities, researcher Jamal Osei et al. surveyed fiddler crabs, marsh periwinkles, and 58 other species at multiple sites along the harbor. Using the Index of Marsh Invertebrate Community Integrity (IMICI), on which a high score corresponds to high community integrity, the researchers found that seawalls are more strongly negatively correlated with marsh invertebrate community integrity than revetments are.

Which finding, if true, would most directly illustrate the researchers' finding?

A) Invertebrate communities at Fort Johnson, a site with a high percentage of shoreline consisting of seawalls and revetments, had lower average IMICI scores than did invertebrate communities at Hobcaw Point, a site with a low percentage of shoreline consisting of seawalls and revetments.

B) Invertebrate communities at Drum Island, a site with a relatively high percentage of shoreline consisting of seawalls, had lower average IMICI scores than did invertebrate communities at Shutes Folly, a site with a relatively high percentage of shoreline consisting of revetments.

C) Invertebrate communities at Fort Johnson, a site with equal percentages of shoreline consisting of seawalls and revetments, had higher average IMICI scores than did invertebrate communities at Shutes Folly, a site with different percentages of shoreline consisting of seawalls and revetments.

D) The difference in average IMICI scores for invertebrate communities at Crab Bank and Drum Island, two sites with a higher percentage of shoreline consisting of seawalls than of revetments, was statistically insignificant.

Question 9

Skill: Command of Evidence (Quantitative) | Difficulty: Medium

San Francisco Bay salt marshes provide vital habitat for numerous shorebird species. Historically, cordgrass (*Spartina foliosa*) was the dominant species, but invasive pickleweed (*Salicornia pacifica*) has proven better adapted to increasing soil salinity, spreading more rapidly and tolerating higher salt concentrations than cordgrass. Although the expansion of pickleweed has been associated with a marked rise in total salt-marsh vegetation cover in the bay, ecologists caution that the overall change does not necessarily make the marsh ecosystem more resistant to environmental disturbances.

[PLACEHOLDER: Line graph titled 'San Francisco Bay Salt-Marsh Vegetation Cover 2014–2021.' X-axis: Year (2014–2021). Y-axis: Vegetation area (hectares) from 0 to 12,000. Three lines: solid triangle = cordgrass (relatively flat around 2,000–2,500 hectares, with a slight dip in 2021); dashed square = pickleweed (rising from ~6,000 in 2014 to ~10,000 in 2019, then a dip to ~8,000 in 2021); dotted circle = total of all marsh vegetation (rising from ~9,000 in 2014 to ~11,500 in 2019, then dropping to ~9,500 in 2021).]

Which statement, if true, would account for data shown in the graph and would illustrate the point made by the ecologists?

A) In early 2021, an unusually severe drought increased soil salinity throughout the bay, leading to conditions that inhibited the growth of most marsh plant species.

B) In early 2020, a fungal pathogen that affects cordgrass but not pickleweed spread through several marsh sites in the bay.

C) Between 2014 and 2019, the total area covered by pickleweed and the total area covered by all marsh vegetation both increased as soil salinity in the bay rose.

D) Soil salinity in the bay increased gradually from 2014 to 2020, but in early 2021 there was an unprecedented salinity spike, reaching levels tolerable by few marsh species other than pickleweed.

Question 10

Skill: Command of Evidence (Quantitative) | Difficulty: Medium

Some wildlife studies estimate how far migratory birds travel annually by fitting individuals with GPS transmitters. Other studies track annual route distance (ARD), which equals twice the straight-line distance between a species' breeding and wintering grounds. A biologist argues that because GPS records all detours an individual takes, using GPS to track a species would yield substantially longer recorded travel distances than using ARD. For example, it is very likely that the distance reported for the ______

[PLACEHOLDER: Table titled 'Reported Annual Travel Distances in Four Studies of Migratory Bird Populations' with columns: Species | Region | Distance (km) | Measurement method. Rows: Bar-tailed godwit | Oceania | 4,600 | ARD; Barn swallow | Europe | 11,200 | GPS; Arctic tern | Arctic | 18,000 | GPS; Osprey | Africa | 6,800 | ARD.]

Which choice most effectively uses data from the table to complete the example?

A) Arctic tern would be less than 18,000 kilometers if the ARD method had been used.
B) barn swallow would be less than 11,200 kilometers if the ARD method had been used.
C) bar-tailed godwit would be greater than 4,600 kilometers if the GPS method had been used.
D) Arctic tern would be greater than 11,200 kilometers if the GPS method had been used.

Question 11

Skill: Inferences | Difficulty: Medium

Thermoelectric generators convert temperature differences (thermal gradients) into electrical energy, eliminating the need for separate power sources. The exhaust system of a diesel truck, for example, can supply enough waste heat to power several of its onboard sensors thermoelectrically. A recently developed thermoelectric module incorporating a high-performance bismuth telluride (Bi_2Te_3) semiconductor has been shown to deliver stable voltage output under steady thermal gradients, an essential requirement for electronic instruments to operate thermoelectrically.

Which finding, if true, would most directly support the text's claim about electronic instruments?

A) The nearly constant thermal gradient across a spacecraft's hull makes it feasible to power its diagnostic instruments using only standard thermoelectric modules without Bi_2Te_3 semiconductors.
B) Irregular or fluctuating voltage supply compromises the accuracy of electronic instruments.
C) The Bi_2Te_3 semiconductor is physically incompatible with most commercially available instrument housings.
D) The high thermal conductivity of Bi_2Te_3 is the property that enables a thermoelectric module to produce voltage output sufficient for electronic instruments.

Question 12

Skill: Inferences | Difficulty: Hard

Retailers increasingly anticipate that merchandise ordered online will arrive within hours, let alone days, of purchase. Although advances in long-haul freight have substantially shortened inter-warehouse transit times, last-mile delivery (the final leg from a local depot to the customer's doorstep) remains a persistent bottleneck. Time pressure driven by consumer expectations is not the sole obstacle; other complications, such as growing traffic congestion in metropolitan areas, also persist. While technological remedies to these challenges have been proposed — the deployment of autonomous ground vehicles, for instance — progress has been hampered by unforeseen practical difficulties (e.g., the absence of standardized curb-access protocols in residential neighborhoods). Consequently, ______

Which choice most logically completes the text?

- A) in the near term, retailers are unlikely to eliminate the obstacles associated with last-mile logistics.
- B) technological solutions for last-mile logistics are expected to raise consumer expectations for faster delivery.
- C) retailers should prioritize investment in established long-haul freight improvements over experimental last-mile innovations.
- D) the deployment of autonomous ground vehicles may help retailers meet consumer expectations today but is probably not sustainable over the long term.

Question 13

Skill: Inferences | Difficulty: Medium

Paleontologists searching for traces of early vertebrate life in Africa have recovered numerous fossils of ancient fish, including specimens from a geological formation at the Miguasha site in Morocco. However, to locate even older specimens such as early jawless fish from the Ordovician period, researchers must investigate sites on other continents, such as the Ordovician-age geological formation at Soom in South Africa, because ______

Which choice most logically completes the text?

- A) Africa has formations containing ancient fish but not formations containing early Ordovician jawless fish.
- B) early jawless fish have been discovered at more sites globally than ancient fish with jaws have.
- C) the Soom formation dates to an earlier geological period than the formation at Miguasha.
- D) the ancient fish at Miguasha are better preserved than the jawless fish at Soom.

Question 14

Skill: Inferences | Difficulty: Hard

Microbial electrolysis cells (MECs) harness the metabolic activity of certain bacteria that oxidize organic compounds, releasing electrons. The bacteria colonize the surface of a carbon-fiber anode, but channeling the electrons from the bacterial cytoplasm to an external circuit requires that the electrons traverse a cascade of thermodynamically unfavorable reduction-oxidation (redox) steps. As a result, MEC current output seldom exceeds a density of 0.8 milliamps per square centimeter (mA/cm^2). In a trial, researchers coated the carbon-fiber anode of an MEC with gold nanoparticles. The resulting current density was 1.5 mA/cm^2. Because metals like gold exhibit high electrical conductivity, the researchers hypothesized that ______

Which choice most logically completes the text?

A) gold nanoparticles may enhance the bacteria's metabolic rate, thereby increasing the number of electrons available for transfer to the circuit.

B) as the bacterial colony grows denser, the cascade of redox steps may accelerate independently of the gold nanoparticles.

C) electrons may be transferred directly to the circuit before the gold nanoparticles facilitate the redox cascade.

D) gold nanoparticles may enable electrons to bypass the cascade of redox steps and transfer directly to the external circuit.

Question 15

Skill: Inferences | Difficulty: Hard

The proportion of carbon dioxide to other atmospheric gases — expressed by a metric called the CO_2 mole fraction — governs a range of climatological processes, particularly global mean temperature and ocean acidification. For Mars, the observational data available are too fragmentary and inconsistent to fully define the range of the CO_2 mole fraction at different altitudes. Lara Chen and colleagues observe that the outputs of the GEM atmospheric model of Mars, which correspond well to observations in certain respects, reflect the model developers' response to this challenge: prescribing a constant CO_2 mole fraction for the lowest layer of the atmosphere. It is therefore important to note that ______

Which choice most logically completes the text?

A) additional observations of Mars may resolve the planet's CO_2 mole fraction precisely enough for the model to use a single value rather than a range.

B) some discrepancies between the model's simulations of Martian surface temperature and the planet's actual surface temperature are to be expected.

C) inconsistencies in the model's simulations of Martian surface temperature could be attributable to variations in the planet's CO_2 mole fraction.

D) even though the model's outputs sometimes match observational data, Mars's actual CO_2 mole fraction is probably higher than the fraction the model uses.

Question 16

Skill: Standard English Conventions | Difficulty: Easy

Roberto Matta, a Chilean-born painter and architect, produced numerous large-scale surrealist canvases, such as his celebrated series *The Earth Is a Man*. In these works, Matta's imaginative vision is exemplified by his use of biomorphic ______ often integrate mathematical curves and cosmic imagery.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) shapes. That
B) shapes: that
C) shapes; that
D) shapes that

Question 17

Skill: Standard English Conventions | Difficulty: Easy

Capoeira, a martial art that originated in Brazil, has gained an international ______ 2018, the Portuguese duo known as Rafael and Mariana triumphed over competitors from around the globe to win the annual World Capoeira Championship held in Salvador, Bahia, becoming the first European winners.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) following in
B) following, in
C) following and in
D) following. In

Question 18

Skill: Standard English Conventions | Difficulty: Easy

The Dravidian language family encompasses roughly 80 languages — including the ______ Tamil and Kannada, which are spoken by 75 million and 44 million people, respectively — and accounts for nearly one-quarter of all languages spoken in South Asia, making it of great interest to linguists like Bhadriraju Krishnamurti.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) languages,
B) languages:
C) languages
D) languages—

Question 19

Skill: Standard English Conventions | Difficulty: Medium

Although 4 Vesta and 1 Ceres are both classified as large bodies in the main asteroid belt — rocky objects in stable orbits between Mars and Jupiter — they exhibit notable differences in ______ object 4 Vesta has a basaltic surface marked by extensive volcanic resurfacing (evidence of a differentiated interior), while the larger body, 1 Ceres, showing no such resurfacing, is considered geologically quiescent.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) geology; the
- B) geology. The
- C) geology, while the
- D) geology: while the

Question 20

Skill: Standard English Conventions | Difficulty: Medium

Chloroplast genomes replicate independently of the nuclear genome, a process that should over many generations lead to an accumulation of deleterious mutations and a reduction in photosynthetic efficiency. However, nuclear-encoded repair enzymes are hypothesized to coevolve with the rate of chloroplast mutation, eliminating damaged sequences. Such a compensatory mechanism ______ the organelle's deterioration, unlike accessory pigment proteins, offers an evolutionary rationale for why chloroplasts have persisted across millions of years of plant evolution.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) had offset
- B) offset
- C) offsets
- D) offsetting

Question 21

Skill: Standard English Conventions | Difficulty: Medium

In a national referendum, one might expect advocacy groups to concentrate their outreach efforts on the most densely populated provinces. However, if polling and historical turnout data indicate that the outcome in a given province is essentially predetermined, an advocacy group will not allocate its resources there. Ultimately, a province's past participation rates and polling trends, not its population size, ______ its strategic value to advocacy campaigns.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) establishes
- B) has established
- C) establishing
- D) establish

Question 22

Skill: Transitions | Difficulty: Easy

To trace subtle developments in the tradition of Japanese woodblock printing, first examine Suzuki Harunobu's "Plum Blossoms at Night" from 1766; ______ compare it to Utagawa Hiroshige's "Plum Estate, Kameido" from 1857.

Which choice completes the text with the most logical transition?

A) therefore,
B) still,
C) instead,
D) next,

Question 23

Skill: Transitions | Difficulty: Medium

Ground contaminated with cadmium (a toxic heavy metal) is hazardous to most vegetation and wildlife, but the plant species *Thlaspi caerulescens*, or alpine pennycress, not only tolerates cadmium-laden soils but actively extracts the metal. As a metal hyperaccumulator, *Thlaspi caerulescens* absorbs significant quantities of cadmium and sequesters it in its leaf tissues; ______ cadmium concentrations in the surrounding soil decline.

Which choice completes the text with the most logical transition?

A) specifically,
B) consequently,
C) nonetheless,
D) furthermore,

Question 24

Skill: Rhetorical Synthesis | Difficulty: Easy

- The French word 'franc' derives from the Latin inscription 'Francorum Rex' (King of the Franks).
- Today, more than a dozen different national currencies include the word 'franc' in their names.
- Rwanda's currency is called the Rwandan franc.
- Rwanda is a landlocked country in East Africa.

The student wants to give an example of a country that uses the word 'franc' in the name of its currency. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Rwanda uses the Rwandan franc as its official currency.
B) Rwanda does not name its currency using the German word 'pfennig.'
C) The French word 'franc' is the etymological root of the English word 'frank.'
D) At present, more than a dozen nations use the word 'franc' in their currency names.

Question 25

Skill: Rhetorical Synthesis | Difficulty: Medium

- Formal analysis and historical analysis are two methods used in art criticism.
- Formal analysis focuses on an artwork's visual elements — line, color, composition — and how they contribute to its overall effect.
- A formal analysis of Georgia O'Keeffe's *Black Iris* might consider how the painting's close-up composition and subtle color gradients shape the viewer's experience.
- Historical analysis examines the social, political, and cultural context in which a work was created.
- A historical analysis of Diego Rivera's *Man at the Crossroads* might consider how the mural's depiction of industrialization reflects the artist's political ideologies.

The student wants to present the concept of historical analysis to an audience unfamiliar with it. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) One method of art criticism, historical analysis examines the social and political circumstances surrounding a work's creation.

B) Diego Rivera's mural *Man at the Crossroads* depicts industrialization's impact and thus reflects the artist's political commitments.

C) A formal analysis of Georgia O'Keeffe's *Black Iris* might explore how the painting's close-up composition and color gradients define its visual impact.

D) Formal analysis and historical analysis offer contrasting lenses because the former concentrates on visual elements while the latter concentrates on context.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Medium

- Carbon can adopt several distinct structural forms called allotropes.
- Diamond is a carbon allotrope.
- Diamond consists of carbon atoms bonded in a rigid three-dimensional tetrahedral lattice.
- Graphene is also a carbon allotrope.
- Graphene consists of carbon atoms arranged in flat, single-atom-thick hexagonal sheets.

The student wants to highlight a structural difference between diamond and graphene. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Arranged in cylindrical tubes, carbon nanotubes differ from diamond, a carbon allotrope with a three-dimensional lattice.

B) Diamond and graphene are both allotropes consisting entirely of carbon atoms, yet carbon can adopt many distinct atomic arrangements.

C) Unlike graphene, in which carbon atoms are arranged in flat hexagonal sheets, diamond consists of carbon atoms bonded in a rigid three-dimensional lattice.

D) The carbon atoms in diamond form a three-dimensional lattice, and the carbon atoms in graphene form flat hexagonal sheets.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Hard

- Learning theory is the study of how organisms acquire new behaviors.
- 1938: psychologist B. F. Skinner formalized operant conditioning, proposing that behaviors are shaped by their consequences (reinforcement and punishment).
- In Skinner's framework, learning requires the organism to perform a behavior and experience its consequences directly.
- 1977: psychologist Albert Bandura introduced social learning theory (SLT).
- SLT holds that organisms can also learn by observing the behaviors and outcomes experienced by others, without direct reinforcement.

The student wants to compare Skinner's framework with Bandura's. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In 1977, Bandura introduced social learning theory, which diverged from Skinner's earlier operant conditioning framework that held the mind was composed of stimulus-response associations.

B) Because Skinner's operant conditioning treats some but not all behavioral processes as learned, it is narrower in scope than Bandura's social learning theory.

C) Both Skinner's and Bandura's frameworks address how organisms acquire new behaviors, but they disagree on whether observation alone can produce learning.

D) Building on Skinner's 1938 operant conditioning framework, Bandura proposed that learning encompasses all behavioral processes.

Section 1: Reading & Writing — Module 2, Harder Path (27 Questions)

Question 1

Skill: Words in Context | Difficulty: Medium

The tidal marshes of the Wadden Sea are not ______: researchers have documented measurable shifts in sediment accretion rates, salinity gradients, and invertebrate community composition across intervals as brief as a single tidal cycle, underscoring the system's responsiveness to even minor hydrological perturbations.

Which choice completes the text with the most logical and precise word or phrase?

A) conspicuous
B) static
C) fragile
D) productive

Question 2

Skill: Words in Context | Difficulty: Hard

Although the biennial Venice Architecture Exhibition routinely showcases speculative designs that challenge structural and material conventions, critic Francesca Moretti contends that many entries remain ______: their conceptual ambitions, however provocative on paper, rely on fabrication techniques that have been standard in parametric architecture for over a decade and therefore contribute little that practicing engineers would regard as genuinely novel.

Which choice completes the text with the most logical and precise word or phrase?

A) derivative
B) hazardous
C) accessible
D) premature

Question 3

Skill: Words in Context | Difficulty: Hard

The proliferation of citizen-science platforms has democratized data collection in ornithology, but the resulting datasets present a methodological ______: because volunteer observers vary widely in expertise and effort, analysts must develop sophisticated statistical corrections before the data can be incorporated into peer-reviewed population estimates.

Which choice completes the text with the most logical and precise word or phrase?

A) innovation
B) complication
C) redundancy
D) incentive

Question 4

Skill: Text Structure and Purpose | Difficulty: Hard

The following passage is adapted from Edith Wharton's 1920 novel *The Age of Innocence*. Newland Archer is attending an opera performance in New York City.

It was a crowded night at the Academy, and as Newland Archer leaned against the back wall of the club box he surveyed the scene with a sense of proprietorship that the habitué of such evenings might have found comical. The fact was that he took New York society entirely for granted: its carefully calibrated hierarchies, its rituals of arrival and departure, its insistence on maintaining a front of perfect composure even while the most turbulent undercurrents ran beneath. He saw nothing to criticize in these arrangements; on the contrary, he regarded them as the natural order and himself as one of their rightful custodians.

Which choice best describes the function of the text within the broader narrative?

A) It establishes the protagonist's unreflective comfort within a social system whose assumptions the novel will later scrutinize.

B) It introduces the protagonist's growing dissatisfaction with social conventions that he considers outdated.

C) It dramatizes a conflict between the protagonist and an antagonist who threatens his standing in elite society.

D) It conveys the protagonist's nostalgia for a vanishing way of life that he knows he cannot preserve.

Question 5

Skill: Text Structure and Purpose | Difficulty: Hard

Economic historian Beatriz Saldaña has argued that the integration of Indigenous weaving cooperatives into global fair-trade supply chains during the 1990s represented a structural shift in how artisan labor was valued internationally. Sociologist Kwame Mensah, drawing on ethnographic fieldwork in Ghana, has complicated this thesis by demonstrating that fair-trade certification often imposed standardization requirements that eroded the very craft traditions it purported to protect. In a 2021 monograph, textile scholar Anya Petrov synthesized both perspectives, proposing a framework that distinguishes between the economic and cultural dimensions of fair-trade participation.

Which choice best describes the overall structure of the text?

A) It presents a claim, introduces a counterargument that challenges that claim, and then describes a study that attempts to reconcile the two positions.

B) It describes two competing methodologies for studying a phenomenon and then evaluates which methodology produces more reliable findings.

C) It introduces a historical event, provides two conflicting explanations for why the event occurred, and then proposes a third explanation.

D) It summarizes a widely accepted theory, describes an experiment that disproved the theory, and then identifies the questions that remain unanswered.

Question 6

Skill: Central Ideas and Details | Difficulty: Hard

The following passage is adapted from a translation of Natsume S■seki's 1914 novel *Kokoro*. The narrator, a young student, is reflecting on his relationship with an older man he calls "Sensei."

I could never decide whether Sensei's habitual reticence concealed a deep reservoir of feeling or merely an emptiness too vast to furnish with words. On those rare occasions when he did speak at length, his observations struck me as penetrating — almost uncomfortably so — yet they were delivered in a tone so detached that I sometimes wondered whether he was addressing me or conducting a solitary audit of his own conclusions. It was this ambiguity, more than any single remark, that bound me to him: I kept returning not because I understood him but because the prospect of understanding him seemed perpetually, tantalizingly close.

Which choice best states the main idea of the text?

A) The narrator is frustrated by Sensei's unwillingness to share personal information and resolves to confront him.

B) The narrator is drawn to Sensei precisely because Sensei remains enigmatic and just beyond full comprehension.

C) The narrator admires Sensei's intellectual brilliance and hopes to emulate his rhetorical style.

D) The narrator suspects that Sensei deliberately cultivates an air of mystery to manipulate those around him.

Question 7

Skill: Cross-Text Connections | Difficulty: Hard

Text 1

Deep-sea hydrothermal vent ecosystems rely on chemosynthetic bacteria that oxidize hydrogen sulfide to produce organic molecules — a process analogous to photosynthesis but independent of sunlight. Marine biologist Ingrid Hagen has proposed that vent-endemic species exhibit limited genetic connectivity between geographically distant vent fields because the intervening abyssal plains represent inhospitable dispersal barriers.

Text 2

Population geneticist Ravi Shankar and colleagues analyzed mitochondrial DNA from tubeworm populations at six hydrothermal vent fields spanning 4,000 kilometers along the Mid-Atlantic Ridge. Contrary to the barrier hypothesis, they found negligible genetic differentiation between populations, suggesting that larval dispersal mechanisms — possibly aided by deep-ocean currents — maintain gene flow across vast distances.

Based on the texts, how would Shankar et al. (Text 2) most likely respond to Hagen's proposal in Text 1?

A) By arguing that chemosynthetic bacteria, rather than tubeworms, are the more appropriate model organism for testing dispersal hypotheses.

B) By presenting evidence that vent-endemic species can traverse the abyssal plains more successfully than Hagen's model predicts.

C) By conceding that genetic connectivity is low but attributing the pattern to selective pressures rather than dispersal barriers.

D) By noting that Hagen's proposal applies only to vent fields in the Pacific and cannot be generalized to the Atlantic.

Question 8

Skill: Inferences | Difficulty: Hard

Developmental psychologist Clara Weiss and colleagues conducted a longitudinal study tracking 300 bilingual children from ages 3 to 7. The researchers hypothesized that children who regularly code-switch — alternate between two languages within a single conversation — would demonstrate superior executive-function skills compared to monolingual peers. The study confirmed this advantage for tasks requiring inhibitory control but, unexpectedly, found no significant difference in working-memory performance between the two groups.

Which finding, if true, would most directly challenge the researchers' original hypothesis as stated?

A) A separate study found that trilingual children outperformed bilingual children on inhibitory-control tasks.

B) The bilingual children in the study who code-switched most frequently showed no measurable advantage on any executive-function task.

C) Working-memory performance among the monolingual children improved steadily from age 3 to age 7.

D) The bilingual children who rarely code-switched scored comparably to those who code-switched frequently on inhibitory-control tasks.

Question 9

Skill: Inferences | Difficulty: Hard

Geochemist Ayumi Nakano has proposed that the anomalously high concentration of iridium found in a clay layer at the Cretaceous-Paleogene (K-Pg) boundary at sites worldwide is best explained by a single large-asteroid impact. Volcanologist Deepak Prasad, however, argues that the Deccan Traps eruptions in present-day India — which spanned roughly 800,000 years straddling the K-Pg boundary — could have released sufficient iridium from Earth's mantle to account for the observed enrichment without invoking an extraterrestrial source. If Prasad's volcanic hypothesis is correct, one would expect that ______

Which choice most logically completes the text?

A) the iridium enrichment at the K-Pg boundary would be confined to sites nearest the Deccan Traps rather than distributed globally.

B) iridium concentrations in the boundary clay would correlate with the temporal pattern of Deccan volcanism, showing elevated levels across a broad stratigraphic interval rather than a single sharp spike.

C) the isotopic signature of boundary-layer iridium would be indistinguishable from that of iridium found in ordinary chondritic meteorites.

D) other platinum-group elements commonly associated with asteroid impacts would be absent from the boundary layer.

Question 10

Skill: Command of Evidence (Quantitative) | Difficulty: Hard

A team of agricultural economists studied the relationship between annual rainfall variability and crop-insurance claim rates in four provinces of a semi-arid country over the period 2010–2022. The researchers found that provinces with higher rainfall variability consistently exhibited higher claim rates, but the strength of this relationship differed by province. The contrast is most clearly illustrated by the data for ______

[PLACEHOLDER: Table titled 'Rainfall Variability and Crop-Insurance Claim Rates, 2010–2022' with columns: Province | Rainfall variability (coefficient of variation, %) | Average annual claim rate (%). Rows: Province A | 18 | 6.2; Province B | 31 | 14.8; Province C | 22 | 8.5; Province D | 45 | 22.1. The contrast between B and D (largest spread in both measures) best illustrates the researchers' finding.]

Which choice most effectively uses data from the table to complete the statement?

- A) Province A and Province C, which had similar claim rates despite differing rainfall variability.
- B) Province B and Province D, which had the largest difference in both rainfall variability and claim rate.
- C) Province A and Province D, which had the smallest difference in claim rate despite having the largest difference in rainfall variability.
- D) Province B and Province C, which had nearly identical rainfall variability but substantially different claim rates.

Question 11

Skill: Command of Evidence (Quantitative) | Difficulty: Hard

A materials scientist is investigating the thermal conductivity of four experimental polymer composites at two temperatures (25°C and 150°C). The scientist notes that Composite P had a thermal conductivity of 0.45 W/m·K at 25°C and ______

[PLACEHOLDER: Table titled 'Thermal Conductivity of Polymer Composites (W/m·K)' with columns: Composite | 25°C | 150°C. Rows: Composite P | 0.45 | 0.62; Composite Q | 0.72 | 0.91; Composite R | 0.38 | 0.55; Composite S | 0.29 | 0.38. The correct completion references Composite P's value at 150°C: 0.62.]

Which choice most effectively uses data from the table to complete the statement?

- A) 0.62 W/m·K at 150°C.
- B) 0.38 W/m·K at 150°C.
- C) 0.91 W/m·K at 150°C.
- D) 0.55 W/m·K at 150°C.

Question 12

Skill: Command of Evidence (Textual) | Difficulty: Hard

"Night Bazaar" is a 2012 poem by Ananya Kapoor. In the poem, the speaker conveys that the marketplace described is animated by the personal histories of the vendors who inhabit it, observing that ______

Which choice most effectively uses a quotation from 'Night Bazaar' to illustrate the claim?

- A) a passing shopper "senses that old Farida / Never copied her spice blends from any recipe at all. / Rather they emerged / Directly from the kitchen of her own memory."
- B) the aromas of the stalls "Drift and overlap / In the cadence of old Farida's gestures, / Drift and overlap."
- C) dim lanterns "sway and flicker / Above Farida's stall."
- D) the vendors begin their work during "Dusk hours beneath the corrugated awning."

Question 13

Skill: Standard English Conventions | Difficulty: Medium

The architect Zaha Hadid, who was born in Baghdad and later established her practice in London, was the first woman to receive the Pritzker Architecture Prize. Her designs, characterized by sweeping curves and fragmented geometric ______ challenged conventional notions of how buildings interact with their surrounding landscapes.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) volumes:
- B) volumes,
- C) volumes;
- D) volumes

Question 14

Skill: Standard English Conventions | Difficulty: Medium

Marine conservation organizations have increasingly advocated for the establishment of large-scale marine protected areas (MPAs) in the Southern Ocean. These organizations argue that MPAs would safeguard critical habitats for species such as Antarctic krill, ______ serve as a baseline for measuring the ecological effects of climate change on polar marine ecosystems.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) and it would also
- B) which would also
- C) also they would
- D) and would also

Question 15

Skill: Standard English Conventions | Difficulty: Hard

Endosymbiotic theory posits that mitochondria originated as free-living prokaryotes that were engulfed by ancestral eukaryotic cells, a process that should over evolutionary time have led to a progressive transfer of genetic material from the organelle to the host nucleus. Recent genomic analyses, however, suggest that certain mitochondrial genes have resisted nuclear transfer, ______ that these genes encode proteins whose function depends on their being translated within the organelle itself.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) implying
- B) implied
- C) to imply
- D) having implied

Question 16

Skill: Standard English Conventions | Difficulty: Hard

Neither the elaborate kinetic sculptures of Alexander Calder nor the monumental steel arcs of Richard Serra ______ without the mid-twentieth-century innovations in industrial welding that made large-scale metal fabrication commercially viable for artists.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) was achievable
- B) were achievable
- C) has been achievable
- D) is achievable

Question 17

Skill: Standard English Conventions | Difficulty: Hard

The genomic data collected by the Earth BioGenome Project, an international consortium that aims to sequence the DNA of every known eukaryotic species on the planet, ______ expected to take at least a decade to compile, annotate, and make publicly accessible.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) are
- B) is
- C) were
- D) have been

Question 18

Skill: Transitions | Difficulty: Medium

Permafrost — permanently frozen ground — underlies vast areas of the Arctic. As global temperatures rise, the upper layers of permafrost thaw, releasing methane and carbon dioxide that had been trapped for millennia. ______ the thawing process also destabilizes the ground surface, causing infrastructure damage in Arctic communities.

Which choice completes the text with the most logical transition?

- A) In contrast,
- B) For example,
- C) Moreover,
- D) Nevertheless,

Question 19

Skill: Transitions | Difficulty: Hard

The discovery of high-temperature superconductors in 1986 initially prompted widespread optimism that lossless electrical transmission would soon become commercially practical. ______ engineering obstacles — particularly the brittleness of the ceramic materials involved and the difficulty of manufacturing them into flexible wires — have so far prevented large-scale deployment.

Which choice completes the text with the most logical transition?

- A) Consequently,
- B) Likewise,
- C) In other words,
- D) However,

Question 20

Skill: Transitions | Difficulty: Hard

The concept of 'urban heat islands' — metropolitan areas that are significantly warmer than surrounding rural regions — is well documented. Less widely appreciated, ______ is the phenomenon of 'urban cool islands,' in which dense tree canopy cover and reflective building materials can lower daytime temperatures in certain city neighborhoods below those recorded in adjacent open farmland.

Which choice completes the text with the most logical transition?

- A) in addition,
- B) as a result,
- C) however,
- D) specifically,

Question 21

Skill: Rhetorical Synthesis | Difficulty: Hard

- Phytochrome is a photoreceptor protein in plants that detects red and far-red light wavelengths.
- Phytochrome regulates photoperiodic responses, including the timing of flowering and seed germination.
- In 2019, botanists Tanaka and Fujimoto demonstrated that modifying phytochrome sensitivity in wheat varieties could alter their flowering schedule by up to three weeks.
- The researchers suggested that this finding could be applied to extend growing seasons in high-latitude agricultural regions where daylight hours are limited during winter.

The student wants to emphasize the practical applications of the research. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) By engineering crops with enhanced phytochrome sensitivity, agricultural scientists could potentially extend the growing season in high-latitude regions where winter daylight is limited.
- B) Phytochrome proteins were first identified in the 1950s by researchers at the United States Department of Agriculture.
- C) The experiments of Tanaka and Fujimoto confirmed earlier predictions about phytochrome's role in photoperiodic responses.
- D) Red and far-red light wavelengths interact differently with phytochrome proteins in plant cells.

Question 22

Skill: Rhetorical Synthesis | Difficulty: Hard

- Art Nouveau was an architectural and decorative movement prominent from the 1890s to roughly 1910.
- Art Nouveau is characterized by elaborate ornamentation, curving organic lines, and motifs drawn from natural forms such as flowers, vines, and insects.
- The Bauhaus was an architectural and design school founded in 1919 in Weimar, Germany.
- Bauhaus design is characterized by geometric simplicity, the rejection of ornamentation, and the use of industrial materials such as steel, glass, and concrete.
- Both movements were influential in shaping twentieth-century architecture and design.

The student wants to emphasize a difference between the two architectural movements. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Both Art Nouveau and the Bauhaus were influential architectural movements of the early twentieth century that attracted practitioners from across Europe.

B) Whereas Art Nouveau embraced elaborate organic ornamentation inspired by natural forms, the Bauhaus favored austere geometric simplicity and industrial materials.

C) The Bauhaus school, established in Weimar in 1919, trained architects who went on to design many of the twentieth century's most iconic buildings.

D) Art Nouveau buildings can be identified by their curving facades and floral motifs, features that distinguish them from buildings of most other periods.

Question 23

Skill: Rhetorical Synthesis | Difficulty: Hard

- In a 2020 study, Park and Gupta examined the bioaccumulation of mercury in freshwater mussels in a river system.
- The study sites were located along a temperate river in the mid-Atlantic United States.
- Some sites had chronically low dissolved-oxygen levels due to upstream agricultural runoff.
- The researchers found that mussels at low-oxygen sites bioaccumulated mercury at rates three times higher than mussels at well-oxygenated sites.
- Bioaccumulation is the process by which organisms absorb and retain contaminants in their tissues faster than they can eliminate them.

The student wants to present the study's findings to an audience already familiar with the concept of bioaccumulation. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The findings of Park and Gupta's study were published in 2020.

B) In a 2020 study by Park and Gupta, mercury was found to bioaccumulate in freshwater mussels, meaning that the organisms retained the metal in their tissues over time.

C) The effects of heavy-metal contamination — including mercury — on freshwater invertebrates have been the focus of numerous ecotoxicology studies.

D) Park and Gupta found that in a temperate river system with low dissolved-oxygen levels, freshwater mussels bioaccumulated mercury at rates three times higher than those observed at well-oxygenated sites.

Question 24

Skill: Rhetorical Synthesis | Difficulty: Hard

- Christopher Okigbo (1932–1967) and Wole Soyinka (b. 1934) are two of the most celebrated Nigerian poets of the twentieth century.
- Okigbo's poetry is notable for its dense, fragmented imagery and elliptical syntax, techniques drawn from European modernist poets such as Ezra Pound and T. S. Eliot.
- Soyinka's poetry draws heavily on the narrative structures, ritual language, and performative traditions of Yoruba oral literature.
- Both poets addressed themes of colonialism, cultural identity, and political conflict in their work.
- Soyinka was awarded the Nobel Prize in Literature in 1986.

The student wants to compare the two poets' approaches to form. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Both Okigbo and Soyinka are regarded as major figures in the canon of postcolonial African literature.

B) While Okigbo structured his poetry around fragmented, elliptical imagery influenced by European modernism, Soyinka drew more heavily on the narrative and performative traditions of Yoruba oral literature.

C) Christopher Okigbo, a Nigerian poet, was killed during the Biafran War in 1967 at the age of 35.

D) Wole Soyinka was awarded the Nobel Prize in Literature in 1986, becoming the first African laureate in that category.

Question 25

Skill: Rhetorical Synthesis | Difficulty: Hard

- The invasive mountain pine beetle (Dendroctonus ponderosae) has expanded its range across western North America since the early 2000s.
- Native bark beetles (Ips pini) are ecologically important decomposers of dead wood.
- In a 2022 study, entomologists García and Lindström surveyed beetle populations in managed and old-growth forests across British Columbia.
- The researchers predicted that the invasive beetle would suppress native beetle populations in all forest types.
- Contrary to their prediction, in old-growth forests the native species maintained its population density despite the presence of the invasive species.

The student wants to emphasize a finding that surprised the researchers. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) García and Lindström expected the invasive beetle to outcompete the native species in all forest types, but in old-growth stands the native species maintained its population density.

B) The invasive mountain pine beetle has spread across much of western North America since the early 2000s.

C) García and Lindström's 2022 study surveyed beetle populations in forests across British Columbia.

D) Native bark beetles play an important ecological role by accelerating the decomposition of dead wood.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Hard

- In a 2023 study, geographer Fatima Hassan and archaeologist Emeka Okafor investigated ancient irrigation systems in the central Saharan desert.
- Hassan used satellite imagery and remote-sensing data to identify subsurface channels invisible at ground level.
- Okafor applied radiocarbon dating to organic residues found within the channels to establish their age.
- The combined approach revealed a network of irrigation channels dating to approximately 5,000 years ago.
- The researchers argued that neither method alone would have been sufficient to both locate and date the channels.

The student wants to highlight the interdisciplinary nature of the research. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) By combining satellite imagery analysis from remote sensing with radiocarbon dating techniques from archaeology, Hassan and Okafor were able to identify and date previously unknown irrigation channels in the Saharan desert.

B) The Saharan desert contains traces of ancient agricultural systems that date back thousands of years.

C) Hassan and Okafor published their findings in the *Journal of Arid Environment Studies* in 2023.

D) Radiocarbon dating is a well-established technique for determining the age of organic materials up to approximately 50,000 years old.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Hard

- Quantum chromodynamics (QCD) is the theoretical framework describing the strong nuclear force.
- QCD predicts that the proton charge radius — a measure of the proton's spatial extent — should be approximately 0.877 femtometers.
- In a 2023 experiment at the CERN Large Hadron Collider, physicist Alexei Ivanov and colleagues measured the proton charge radius at 0.831 femtometers.
- The 5.2% discrepancy between the predicted and measured values, given the high precision of both, represents a statistically significant tension.
- The researchers suggested that the discrepancy may point to physics beyond the Standard Model.

The student wants to emphasize the scale of the discrepancy between theoretical predictions and experimental results. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Quantum chromodynamics (QCD) is the theoretical framework that describes the strong nuclear force binding quarks within protons and neutrons.

B) The 2023 measurements by Ivanov and colleagues were conducted at the CERN Large Hadron Collider.

C) While QCD predicted a proton charge radius of 0.877 femtometers, Ivanov and colleagues measured it at 0.831 femtometers — a 5.2% discrepancy that, given the precision of both the theory and the measurement, represents a statistically significant tension.

D) The proton charge radius has been the subject of measurement campaigns at several particle accelerators around the world.

Section 2: Math — Module 1 (22 Questions)

Question 1

Skill: Algebra — Linear Equations in One Variable | Difficulty: Easy

The width of a rectangle is 6 centimeters. The length of the rectangle is 40 centimeters longer than the width. What is the area, in square centimeters, of this rectangle?

A) 12
B) 24
C) 92
D) 276

Question 2

Skill: Algebra — Linear Equations in One Variable | Difficulty: Easy

A marine biologist uses a linear model to estimate the length of a juvenile hammerhead shark after it is born. The model estimates that a particular hammerhead shark measures 42 centimeters at birth and grows at a rate of 1.8 centimeters per day for the first 90 days. Based on this model, what is the estimated length, in centimeters, of this shark 10 days after it is born?

(Student-produced response — enter a numerical value)

Question 3

Skill: Algebra — Linear Equations in Two Variables | Difficulty: Easy

[PLACEHOLDER: Scatterplot with u on the x-axis (0 to 8) and v on the y-axis (0 to 18). Approximately 12 data points showing a positive linear trend. The line of best fit has a y-intercept near –1.2 and a slope of approximately 2.0, passing through roughly (1, 1), (3, 5), (5, 9), (7, 13).]

The scatterplot shows the relationship between two variables, *u* and *v*. A line of best fit is also shown.

A) v = 1.2 + 2.0u
B) v = 1.2 – 2.0u
C) v = –1.2 + 2.0u
D) v = –1.2 – 2.0u

Question 4

Skill: Algebra — Systems of Equations | Difficulty: Medium

$y = –0.8$
$y = x^2 + 6x + a$

In the given system of equations, *a* is a positive integer constant. The system has no real solutions. What is the least possible value of *a*?

(Student-produced response — enter a numerical value)

Question 5

Skill: Algebra — Linear Inequalities | Difficulty: Medium

The function *f* is defined by f(x) = 4x – 3. What is the y-intercept of the graph of y = f(x) in the xy-plane?

A) (0, –4)
B) (0, –3)
C) (0, 3)
D) (0, 4)

Question 6

Skill: Algebra — Linear Functions | Difficulty: Easy

If 7(x + 3) = 6(x + 3) + 52, what is the value of x + 3?

A) –3
B) 49
C) 52
D) 55

Question 7

Skill: Algebra — Interpreting Linear Models | Difficulty: Easy

In one week in 2019, a freelance editor earned a total of $845 by working on two projects. She charged $25 per hour on Project A and $18 per hour on Project B. The equation 25a + 18b = 845 models this situation, where *a* is the number of hours on Project A and *b* is the number of hours on Project B. Which of the following is the best interpretation of 25 in this context?

A) The amount, in dollars, the editor earned for each hour she worked on Project A
B) The amount, in dollars, the editor earned for each hour she worked on Project B
C) The number of hours the editor worked on Project A during the week
D) The number of hours the editor worked on Project B during the week

Question 8

Skill: Problem Solving and Data Analysis — Ratios and Proportions | Difficulty: Easy

In 12 days, a migratory goose consumed 51.6 pounds of grain. Which equation describes the amount of grain y, in pounds, the goose consumed in these 12 days?

A) $y = 12$
B) $y = 51.6/12$
C) $y = 51.6$
D) $y = 51.6 + 12$

Question 9

Skill: Problem Solving and Data Analysis — Unit Conversion | Difficulty: Medium

To measure fluctuations in sap pressure, samples of bark were taken from 30 trees and cut in the shape of a cylinder. The diameter of one of these cylinders is 2.000 centimeters and its height is 4.000 centimeters. This cylinder has a density of 0.350 grams per cubic centimeter. What is the mass of this cylinder, in grams? (Use $\pi \approx 3.14159$)

(Student-produced response — enter a numerical value)

Question 10

Skill: Advanced Math — Equivalent Expressions | Difficulty: Easy

The expression $5x^8 + 11x^8 – 3x^8$ is equivalent to bx^8, where b is a constant. What is the value of b?

(Student-produced response — enter a numerical value)

Question 11

Skill: Advanced Math — Exponents and Radicals | Difficulty: Medium

Which expression is equivalent to $(246y)^{1/2}$, where $y > 1$?

A) $246 \cdot \sqrt{y}$
B) $\sqrt{246} \cdot y$
C) $\sqrt{(246y)}$
D) $\sqrt{((246y)^2)}$

Question 12

Skill: Problem Solving and Data Analysis — Probability | Difficulty: Medium

At a trade fair, there are a total of 320 attendees. Each attendee is in either Hall X, Hall Y, or Hall Z. If one attendee is selected at random, the probability of selecting someone in Hall X is 0.55, and the probability of selecting someone in Hall Y is 0.30. How many attendees are in Hall Z?

A) 16
B) 48
C) 96
D) 176

Question 13

Skill: Geometry and Trigonometry — Perpendicular Lines | Difficulty: Medium

Line *p* is defined by $y = -(1/3)x + 12$. Line *q* is perpendicular to line *p* in the xy-plane. What is the slope of line *q*?

A) 12
B) 3
C) 1/3
D) 1/12

Question 14

Skill: Advanced Math — Exponential Functions | Difficulty: Medium

$f(x) = -8(3)^{x/4}$

Which table gives four values of *x* and their corresponding values of f(x) for the given exponential function?

A) x: –4, 0, 4, 8 → f(x): –(8/3), –8, –24, –72
B) x: –4, 0, 4, 8 → f(x): –(8/3), 0, –24, –72
C) x: –4, 0, 4, 8 → f(x): 8/3, –8, –24, –72
D) x: –4, 0, 4, 8 → f(x): –(8/3), –8, –24, –48

Question 15

Skill: Advanced Math — Polynomial Expressions | Difficulty: Easy

Which expression is equivalent to $12(x + 8)$?

A) $12x + 8$
B) $12x + 96$
C) $12x + 20$
D) $12x + 4$

Question 16

Skill: Advanced Math — Function Notation | Difficulty: Medium

The function *g* is defined by $g(x) = 3x + 5$, and the function *h* is defined by $h(x) = 7 - x$. What is the value of $2g(4) - h(4)$?

(Student-produced response — enter a numerical value)

Question 17

Skill: Problem Solving and Data Analysis — Percentages | Difficulty: Medium

A school is purchasing laptop computers for its teachers. The laptops cost $210 each, and a 6% sales tax is applied to the total cost of the order. If the school can spend no more than $48,000 on the laptops, including tax, what is the maximum number of laptops the school can purchase?

(Student-produced response — enter a numerical value)

Question 18

Skill: Geometry and Trigonometry — Angle Conversion | Difficulty: Medium

The difference between the measure of angle *C* and the measure of angle *D* is $-(7/10)\pi$ radians. Which expression shows the difference between the measure of angle *C* and the measure of angle *D*, in degrees?

A) $-(7/10)\pi \cdot (180°/\pi)$
B) $-(7/10)\pi \cdot (\pi/180°)$
C) $-(7/10)\pi \cdot (180°/\pi)$
D) $-(7/10)\pi \cdot (360°/\pi)$

Question 19

Skill: Advanced Math — Exponential Decay | Difficulty: Hard

A chemist observes a sample of a radioactive isotope. An exponential model estimates that the mass, in grams, of the sample decreases by 18% every 31.5 minutes. Which of the following equations could represent this model, where *M* is the estimated mass, in grams, *t* minutes after observation began, and the initial mass is 200 grams?

A) $M = 200(0.18)^{t+31.5}$
B) $M = 200(0.18)^{t/31.5}$
C) $M = 200(0.82)^{t+31.5}$
D) $M = 200(0.82)^{t/31.5}$

Question 20

Skill: Geometry and Trigonometry — Triangles | Difficulty: Hard

In triangle *DEF*, the measure of angle *D* is 42°, the measure of angle *E* is y°, and the measure of angle *F* is $(3y - 6)$°. Point *G* lies on segment *DE*, point *H* lies on segment *EF*, and segment *GH* is parallel to segment *DF*. What is the measure, in degrees, of angle *EHG*? (Disregard the degree symbol when entering your answer.)

(Student-produced response — enter a numerical value)

Question 21

Skill: Geometry and Trigonometry — Triangles and Circles | Difficulty: Hard

The perimeter of an equilateral triangle is 960 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $k\sqrt{3}$ centimeters. What is the value of *k*?

(Student-produced response — enter a numerical value)

Question 22

Skill: Advanced Math — Quadratic Functions | Difficulty: Hard

The function *f* is a quadratic function. In the xy-plane, the graph of $y = f(x)$ has a vertex at $(3, 10)$ and passes through the points $(4, 56)$ and $(1, 56)$. What is the value of $f(-1) - f(0)$?

- A) 230
- B) 322
- C) 506
- D) 598

Section 2: Math — Module 2, Easier Path (22 Questions)

Question 1

Skill: Algebra — Interpreting Linear Models | Difficulty: Easy

In one month in 2021, a bakery spent a total of $936 on two types of flour. Bread flour cost $24 per bag, and pastry flour cost $18 per bag. The equation $24b + 18p = 936$ represents this situation, where b is the number of bags of bread flour and p is the number of bags of pastry flour. Which of the following is the best interpretation of 24 in this context?

A) The cost, in dollars, of each bag of bread flour
B) The cost, in dollars, of each bag of pastry flour
C) The number of bags of bread flour purchased during the month
D) The number of bags of pastry flour purchased during the month

Question 2

Skill: Problem Solving and Data Analysis — Linear Models | Difficulty: Easy

A marine biologist uses a linear model to estimate the weight of a blue whale after it is born. The model estimates that a certain blue whale weighs 6,200 pounds at birth and gains 12.5 pounds per hour for 100 hours after it is born. Based on this model, what is the estimated weight, in pounds, of this blue whale 9 hours after it is born?

(Student-produced response — enter a numerical value)

Question 3

Skill: Algebra — Solving Linear Equations | Difficulty: Easy

$q - 28r = s$

The given equation relates the positive numbers q, r, and s. Which equation correctly expresses q in terms of r and s?

A) $q = s - 28r$
B) $q = s + 28r$
C) $q = 28rs$
D) $q = s/(28r)$

Question 4

Skill: Algebra — Linear Inequalities | Difficulty: Medium

[PLACEHOLDER: Graph in the xy-plane showing a solid line passing through approximately (0, 7) and (3.5, 0) with a negative slope of –2. The region below and to the left of the line is shaded (including the line). The line equation is approximately $y \leq -2x + 7$. The point (5, –2) is in the shaded region.]

The shaded region shown in the graph represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

- A) (–4, 0)
- B) (0, 4)
- C) (0, –4)
- D) (5, –2)

Question 5

Skill: Algebra — Interpreting Functions | Difficulty: Easy

A rectangle has a length that is 36 times its width. The function $y = (36w)(w)$ represents this situation, where *y* is the area, in square meters, of the rectangle and $w > 0$. Which of the following is the best interpretation of $36w$ in this context?

- A) The length of the rectangle, in meters
- B) The area of the rectangle, in square meters
- C) The difference between the length and the width of the rectangle, in meters
- D) The width of the rectangle, in meters

Question 6

Skill: Algebra — Solving Linear Equations | Difficulty: Easy

If $8(x + 5) = 7(x + 5) + 41$, what is the value of $x + 5$?

- A) –5
- B) 36
- C) 41
- D) 46

Question 7

Skill: Algebra — Systems of Equations | Difficulty: Medium

$x + 14 = -7y + 8$
$x - 14 = 7y + 3$

The solution to the given system of equations is (x, y). What is the value of $2x$?

- A) –(7/5)
- B) 11
- C) 22
- D) 25

Question 8

Skill: Problem Solving and Data Analysis — Proportions | Difficulty: Easy

In 10 days, a brown bear consumed 42.5 pounds of salmon. Which equation describes the total amount of salmon *y*, in pounds, the brown bear consumed in these 10 days?

A) y = 10
B) y = 42.5/10
C) y = 42.5
D) y = 42.5 + 10

Question 9

Skill: Problem Solving and Data Analysis — Volume and Density | Difficulty: Medium

To study the buoyancy of different woods, samples were cut into cubes. The edge length of one cube is 5.000 centimeters. This cube has a density of 0.180 grams per cubic centimeter. What is the mass of this cube, in grams?

(Student-produced response — enter a numerical value)

Question 10

Skill: Algebra — Absolute Value Equations | Difficulty: Medium

–3|4x + 8| + 5 = –19

What are all solutions to the given equation?

A) 0
B) 0 and –4
C) 0 and –(16/4)
D) There is no solution.

Question 11

Skill: Algebra — Linear Functions | Difficulty: Medium

For the linear function *p*, p(d) = –4, where *d* is a constant, p(3) = 32, and the slope of the graph of y = p(x) in the xy-plane is 6. For the linear function *t*, t(d) = –4 and t(7) = 56. What is the slope of the graph of y = t(x) in the xy-plane?

A) –2
B) 6
C) 8
D) 10

Question 12

Skill: Algebra — Linear Relationships (Graph) | Difficulty: Medium

[PLACEHOLDER: Graph in the xy-plane from x = –10 to 10, y = –10 to 10. A line with positive slope passes through approximately (–3, –10) and (0, 2.4) and continues upward. The line has a y-intercept of 2.4 and an x-intercept of approximately –0.6. Points are plotted along the line at integer x-values.]

The graph shows a linear relationship between *x* and *y*. Which equation represents this relationship, where *K* is a positive constant?

- A) Kx + 10y = 24
- B) Kx – 10y = –24
- C) 10x + Ky = 24
- D) 10x – Ky = 24

Question 13

Skill: Algebra — Systems of Inequalities (Graph) | Difficulty: Hard

[PLACEHOLDER: Graph in the xy-plane showing a solid line passing through (–12, 0) and (0, –6) with slope = –1/2. The region above the line (including the line) is shaded. The line equation in standard form is –4x – 8y = 48, so r = –4, t = –8, and r + t = –12. Wait, let me recalculate. If the line passes through (–12, 0) and (0, –6): slope = (–6 – 0)/(0 – (–12)) = –6/12 = –1/2. Equation: y = –(1/2)x – 6, or x + 2y = –12. Multiply by 4: 4x + 8y = –48. Since the region is above/left (shaded includes origin test: 0 + 0 = 0 ≥ –48, yes). So rx + ty ≥ –48 with the line being rx + ty = –48. If the line passes through (–8, 0) and (0, –6): slope = –6/8 = –3/4. Equation: y = –(3/4)x – 6, or 3x + 4y = –24. Multiply by 2: 6x + 8y = –48, r = 6, t = 8, r + t = 14. Let me just pick clean values. Line through (–6, 0) and (0, –8): slope = –8/6 = –4/3. 8x + 6y = –48. r + t = 14. Let me simplify: line through (–12, 0) and (0, –4): slope = –4/12 = –1/3. y = –(1/3)x – 4, or x + 3y = –12. Multiply by 4: 4x + 12y = –48. r + t = 16. This is getting complicated. Let me just use: line through (–8, 0) and (0, –6), r = 6, t = 8, r+t = 14. No, let me just pick simple ones. Use line: 4x + 8y = –48. Passes through (–12, 0) and (0, –6). r = 4, t = 8, r + t = 12. Hmm, none of my answer choices work. Let me redesign: line through (–8, 0) and (0, 6): slope = 6/8 = 3/4. y = (3/4)x + 6, or –3x + 4y = 24, or 3x – 4y = –24. Multiply by 2: 6x – 8y = –48. r + t = 6 + (–8) = –2. That works with choice C) –2.]

The shaded region shown represents the solutions to rx + ty ≥ –48, where *r* and *t* are constants. What is the value of r + t?

- A) 6
- B) 2
- C) –2
- D) –6

Question 14

Skill: Geometry and Trigonometry — Surface Area | Difficulty: Hard

The edge length, in centimeters, of cube B is (5/72) the edge length, in centimeters, of cube A. The surface area, in square centimeters, of cube B is *n* times the surface area, in square centimeters, of cube A. What is the value of *n*?

- A) 25/5,184
- B) 125/373,248
- C) 5/72
- D) 25/72

Question 15

Skill: Advanced Math — Exponential Functions (Graph) | Difficulty: Hard

[PLACEHOLDER: Graph in the xy-plane, x from –1 to 5, y from –6 to 12. A decreasing exponential curve y = f(x) + 4 is shown. It passes through approximately (0, 3) and decreases rapidly toward the right. At x = 0, y = f(0) + 4 = 3, so f(0) = –1. Since f(x) = –5^x + C, f(0) = –1 + C = –1 means C could be 0... Let me reconsider. If the graph of y = f(x) + 4 passes through (0, 3), then f(0) + 4 = 3, f(0) = –1. If f(x) = –5^x + C, then f(0) = –1 + C = –1, so C = 0... None of my choices have C = 0. Let me adjust: graph passes through (0, 6). Then f(0) = 2. f(x) = –5^x + C, f(0) = –1 + C = 2, C = 3. Still doesn't match. Let me use: graph passes through (0, 7), so f(0) = 3. –1 + C = 3, C = 4. So f(x) = –5^x + 4. Answer B.]

The graph of y = f(x) + 4 is shown. Which equation defines function *f*?

A) $f(x) = –5^x + 2$
B) $f(x) = –5^x + 4$
C) $f(x) = –5^x + 6$
D) $f(x) = –5^x + 8$

Question 16

Skill: Advanced Math — Distributive Property | Difficulty: Easy

Which expression is equivalent to 18(x + 7)?

A) 18x + 7
B) 18x + 126
C) 18x + 25
D) 18x + 11

Question 17

Skill: Problem Solving and Data Analysis — Proportional Reasoning | Difficulty: Hard

A square map has a side length of 28 inches, and 1 inch on the map represents an actual distance of 15 miles. A smaller version of the same map is printed as a square with the side length 60% shorter than the side length of the previous map. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

A) 6.00
B) 9.00
C) 25.00
D) 37.50

Question 18

Skill: Advanced Math — Systems of Equations (Quadratic-Linear) | Difficulty: Hard

$y = x - d$
$y = -4(x - 10)^2$

In the given system of equations, *d* is a constant. The system has two distinct real solutions. Which of the following could be the value of *d*?

A) 5
B) 9
C) 241/16
D) 18

Question 19

Skill: Geometry and Trigonometry — Triangle Similarity | Difficulty: Hard

In triangle *GHI*, the measure of angle *G* is 62° and GI = 45. In triangle *JKL*, the measure of angle *J* is 62° and JL = 135. Which additional piece of information is sufficient to prove that triangle *GHI* is similar to triangle *JKL*?

A) GH = 50 and JK = 50.
B) GH = 50 and KL = 150.
C) The measures of angle *H* and angle *K* are 44° and 74°, respectively.
D) The measures of angle *H* and angle *K* are 62° and 44°, respectively.

Question 20

Skill: Geometry and Trigonometry — Circumscribed Circle | Difficulty: Hard

The perimeter of an equilateral triangle is 720 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $m\sqrt{3}$ centimeters. What is the value of *m*?

(Student-produced response — enter a numerical value)

Question 21

Skill: Advanced Math — Function Operations | Difficulty: Hard

The function *f* is defined by $f(x) = (x/a) - 20$, where $a < 0$. What is the product of f(14a) and f(6a)?

(Student-produced response — enter a numerical value)

Question 22

Skill: Advanced Math — Exponential Functions | Difficulty: Hard

The function *f* is defined by $f(x) = a \cdot b^{x/n}$, where *a*, *b*, and *n* are constants, and *b* and *n* are integers. If f(3) = 10 and f(6) = 250, what is the value of f(9)?

(Student-produced response — enter a numerical value)

Section 2: Math — Module 2, Harder Path (22 Questions)

Question 1

Skill: Algebra — Interpreting Linear Models | Difficulty: Medium

A contractor charges a flat fee plus an hourly rate for home inspections. For a 3-hour inspection the total charge is $340, and for a 7-hour inspection the total charge is $580. Which of the following represents the contractor's flat fee, in dollars?

A) 60
B) 100
C) 160
D) 240

Question 2

Skill: Algebra — Systems of Equations | Difficulty: Medium

$3x + 5y = 26$
$7x – 5y = –6$

The solution to the given system of equations is (x, y). What is the value of $4x$?

A) 2
B) 8
C) 10
D) 20

Question 3

Skill: Algebra — Quadratic Equations | Difficulty: Medium

$x^2 – 14x + 40 = 0$

What are the solutions to the given equation?

A) $x = 4$ and $x = 10$
B) $x = –4$ and $x = –10$
C) $x = 2$ and $x = 20$
D) $x = –2$ and $x = –20$

Question 4

Skill: Advanced Math — Rational Expressions | Difficulty: Hard

If $(3x^2 + 14x - 5) / (x + 5) = ax + b$ for all $x \neq -5$, what is the value of $a - b$?

- A) 2
- B) 4
- C) 6
- D) 8

Question 5

Skill: Problem Solving and Data Analysis — Statistics | Difficulty: Medium

A data set consists of 120 values, and the mean of the data set is 85. If 15 is added to each of the values in the data set, which of the following gives the mean of the resulting data set?

- A) 70
- B) 85
- C) 100
- D) 1,800

Question 6

Skill: Advanced Math — Polynomial Functions | Difficulty: Hard

The function *g* is defined by $g(x) = (x - 7)(x + 3)(2x - 1)$. In the xy-plane, the graph of $y = g(x)$ crosses the x-axis at three points. What is the sum of the x-coordinates of these three points?

- A) –3.5
- B) 4.5
- C) 5.5
- D) 10.5

Question 7

Skill: Geometry and Trigonometry — Circle Equations | Difficulty: Hard

$(x - 4)^2 + (y + 9)^2 = 64$

In the xy-plane, the graph of the given equation is a circle. What is the length of the diameter of this circle?

- A) 8
- B) 16
- C) 32
- D) 64

Question 8

Skill: Problem Solving and Data Analysis — Scatterplot Interpretation | Difficulty: Medium

A researcher collected data on the number of hours studied and the corresponding test score for 15 students. A line of best fit for the data has a slope of approximately 6.5 and a y-intercept of approximately 42. According to this model, what is the predicted test score for a student who studies for 5 hours?

A) 48.5
B) 74.5
C) 80.0
D) 98.5

Question 9

Skill: Advanced Math — Exponential Growth | Difficulty: Hard

A population of bacteria doubles every 4.5 hours. If the initial population is 500, which equation gives the population *P* after *t* hours?

A) $P = 500(2)^{4.5t}$
B) $P = 500(2)^{t/4.5}$
C) $P = 500(0.5)^{t/4.5}$
D) $P = 500(4.5)^{2t}$

Question 10

Skill: Algebra — Absolute Value | Difficulty: Hard

$-5|3x + 9| + 7 = -38$

What are all solutions to the given equation?

A) 0
B) 0 and –6
C) 0 and –(18/3)
D) There is no solution.

Question 11

Skill: Advanced Math — Function Composition | Difficulty: Hard

Let $f(x) = 2x - 3$ and $g(x) = x^2 + 1$. What is the value of $f(g(4)) - g(f(4))$?

A) –8
B) 0
C) 8
D) 12

Question 12

Skill: Geometry and Trigonometry — Trigonometric Ratios | Difficulty: Hard

In right triangle *PQR*, angle *Q* is the right angle. If sin(P) = 5/13, what is the value of cos(R)?

- A) 5/13
- B) 12/13
- C) 5/12
- D) 13/5

Question 13

Skill: Problem Solving and Data Analysis — Percentages | Difficulty: Medium

A store sells a jacket for $120, which represents a 25% markup over the store's cost. What was the store's cost for the jacket?

- A) $30
- B) $90
- C) $96
- D) $150

Question 14

Skill: Advanced Math — Quadratic Functions | Difficulty: Hard

The function *h* is defined by $h(x) = -2(x - 5)^2 + 18$. What is the maximum value of h(x)?

- A) –2
- B) 5
- C) 18
- D) 68

Question 15

Skill: Algebra — Rational Equations | Difficulty: Hard

For what value of *x* does (2x + 6)/(x – 3) = 4?

- A) 3
- B) 6
- C) 9
- D) 12

Question 16

Skill: Problem Solving and Data Analysis — Probability | Difficulty: Hard

A jar contains 8 red marbles, 12 blue marbles, and 5 green marbles. If two marbles are selected at random without replacement, what is the probability that both marbles are blue?

A) 144/625
B) 12/25
C) 11/50
D) 132/600

Question 17

Skill: Geometry and Trigonometry — Volume | Difficulty: Hard

A right circular cylinder has a volume of $1{,}350\pi$ cubic centimeters and a height of 6 centimeters. What is the radius, in centimeters, of the cylinder?

(Student-produced response — enter a numerical value)

Question 18

Skill: Advanced Math — Systems of Equations (Quadratic-Linear) | Difficulty: Hard

$y = 2x + k$
$y = -3(x - 8)^2$

In the given system, *k* is a constant. For what value of *k* does the system have exactly one real solution?

(Student-produced response — enter a numerical value)

Question 19

Skill: Advanced Math — Polynomial Division | Difficulty: Hard

When the polynomial $4x^3 - 7x^2 + 5x - 12$ is divided by $(x - 2)$, what is the remainder?

A) –6
B) 2
C) 6
D) 10

Question 20

Skill: Geometry and Trigonometry — Coordinate Geometry | Difficulty: Hard

In the xy-plane, the points (2, 5) and (8, 17) lie on line *m*. Line *n* is perpendicular to line *m* and passes through the midpoint of the segment connecting (2, 5) and (8, 17). What is the y-intercept of line *n*?

(Student-produced response — enter a numerical value)

Question 21

Skill: Advanced Math — Function Transformations | Difficulty: Hard

The function *f* is defined by $f(x) = a(x - h)^2 + k$, where a, h, and k are constants. In the xy-plane, the graph of $y = f(x)$ has a vertex at (3, –4) and passes through (0, 14). What is the value of a?

(Student-produced response — enter a numerical value)

Question 22

Skill: Advanced Math — Exponential Functions | Difficulty: Hard

The function *g* is defined by $g(x) = c \cdot d^{x/m}$, where c, d, and m are positive constants with $d \neq 1$. If g(4) = 12 and g(8) = 108, what is the value of g(12)?

(Student-produced response — enter a numerical value)

Question Count Verification

Section	Module	Count
S1: Reading & Writing	Module 1	27
S1: Reading & Writing	Module 2 — Easier	27
S1: Reading & Writing	Module 2 — Harder	27
S2: Math	Module 1	22
S2: Math	Module 2 — Easier	22
S2: Math	Module 2 — Harder	22
	TOTAL	**147**

Easier Module 2: ~70% Easy/Medium, ~30% Hard
Harder Module 2: ~30% Easy/Medium, ~70% Hard